DESCRIPTION OF COURSEWORK

Course Code	EEE312
Course Name	Programming Practices
Lecturer	Hamid Tahaei
Academic Session	2022/09
Assessment Title	Assignment 1

A. Introduction/ Situation/ Background Information

This assignment applies the programming techniques learned on solving linear equation.

B. Course Learning Outcomes (CLO) covered

At the end of this assessment, students are able to:

CLO 1 Design program using advanced algorithms and techniques

CLO 2 Use modern programming techniques to solve scientific problems

C. University Policy on Academic Misconduct

1. Academic misconduct is a serious offense in Xiamen University Malaysia. It can be defined as any of the following:
   i. **Plagiarism** is submitting or presenting someone else's work, words, ideas, data or information as your own intentionally or unintentionally. This includes incorporating published and unpublished material, whether in manuscript, printed or electronic form into your work without acknowledging the source (the person and the work).
   ii. **Collusion** is two or more people collaborating on a piece of work (in part or whole) which is intended to be wholly individual and passed it off as own individual work.
   iii. **Cheating** is an act of dishonesty or fraud in order to gain an unfair advantage in an assessment. This includes using or attempting to use, or assisting another to use materials

that are prohibited or inappropriate, commissioning work from a third party, falsifying data, or breaching any examination rules.

2. All the assessment submitted must be the outcome of the student. Any form of academic misconduct is a serious offense which will be penalised by being given a zero mark for the entire assessment in question or part of the assessment in question. If there is more than one guilty party as in the case of collusion, both you and your collusion partner(s) will be subjected to the same penalty.

D. Instruction to Students

1. This is an individual assignment.

2. Answer the questions in **Section B Tasks**.

3. Submission deadline is 14 Nov 2022.

4. Submit a zipfile <your_student_ID> including the c source code file for each task. Also, Submit the answer cover page paper with your signature and screen shot of your code for each question and the output. Rename the assignment cover page to "<your_student_ID>.docx". A Written report is required to provide explanation for each block/ line of code using the comment operator such as // in the C file. You are required to include the executable file into your zipfile.

E. Evaluation Breakdown

No.	Component Title	Percentage (%)
1.	Report	7
2.	C Code	8
	TOTAL	**15**

F. Task(s)

Question 1

Write a C program to perform the followings for the given matrix *A*:

a. Take an $N \times N$ matrix from input. (e.g. $N= 3$)

b. Check if the given matrix *A* is a symmetric matrix.

*Hint: A square matrix A= $[a_{ij}]$ is said to be symmetric if A = A^T or $a_{ij} = a_{ji}$

*Note: If the matrix is not symmetric, the program should prompt then accurate error message.

[4 marks]

c. Check if the matrix *A* is a Positive Definite Matrix.

*Hint: A matrix is positive definite if all conditions as stated below are satisfied:

- **A** is symmetric, that is, $a_{ij} = a_{ji}$ for $i,j= 1, 2, \ldots, n$.
- All the diagonal elements in **A** are positive, or $a_{ii} > 0$ for $i= 1, 2, \ldots, n$.
- The largest absolute value in each row of **A** is the diagonal element.
- $a_{ii}\, a_{jj} > {a_{ij}}^2$ for $i, j= 1, 2, \ldots, n$ *and* $i \neq j$

*Note: If the matrix is not symmetric, the program should prompt then accurate error message.

*Note: The program should able to take matrices with any given dimensions.

[4 marks]

$$A= \begin{bmatrix} 2 & 1 & 0 \\ 1 & -2 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

The output should look like as follows:

```
Enter a 3x3 matrix
2 1 0
1 -2 1
0 1 0


The matrix A is not a symmetric matrix because, the diagonal element at A[1][1] is not a positive value
The matrix A is not a symmetric matrix because, the diagonal element at A[2][2] is not a positive value
The matrix A is not a symmetric matrix because, the largest absolute value in each row of A is no the diagonal element.
The matrix A is not a symmetric matrix because, the largest absolute value in each row of A is no the diagonal element.
```

*Write explanation in front of each line of code using the comment operator such as "//" if they are complex.

$$A=\begin{bmatrix}2 & 1 & 0\\ 1 & 2 & 1\\ 0 & 1 & 2\end{bmatrix}$$

The output should look like as follows:

```
Enter a 3x3 matrix
2 1 0
1 2 1
0 1 2


The matrix A is a symmetric matrix and a positive definite matrix


Process returned 0 (0x0)   execution time : 0.809 s
Press any key to continue.
```

*Write explanation in front of each line of code using the comment operator such as "//" if they are complex.

APPENDIX 1

MARKING RUBRICS

Component Title	Assignment 1				Percentage (%)	15	
Criteria	Score and Descriptors					Weight (%)	Marks
	Excellent (7)	Good (5-6)	Average (4-5)	Need Improvement (2-3)	Poor (0-1)		
Short Report	Very detailed description of program flow, algorithm and computational aspects of the problem. Excellent explanation of numerical methods applied to solve the engineering problem.	Good description of program flow and algorithm used. Able to explain numerical methods to solve the engineering problem.	Sufficient information on program flow but lacking details description. Able to explain numerical methods used to solve the problem but not in details.	Missing important information on program flow and algorithm. Descriptions contain minor errors. Explanation of numerical solutions is not sufficient.	Not sufficient information on the program flow and algorithm used. Serious technical error in the description of the program. Unable to explain numerical methods applied to the problem.	7	
Criteria	**Excellent (7)**	**Good (5-6)**	**Average (4-5)**	**Need improvement (2-3)**	**Poor (0-1)**		
C Code	Program works According to Given requirements of The questions. Able to apply Numerical methods learned to solve the problem. Creative use of algorithm and/or advanced programming techniques to improve computational efficiency. Use good programming practices and Code optimization for writing easy to understanding code and efficiency.	Program works according to the given the requirements of the question. Able to apply numerical methods learned to solve the problem. Organization of the code can be improved to simplify understanding and improve execution efficiency.	Program works but only fulfilling partial requirements of the questions. Able to apply numerical methods learned to solve the problem. Code is difficult to understand, not optimized for efficiency.	Program works but results are not correct. Only part of the requirements of the question are fulfilled. Unable to apply numerical methods learned to solve the problem.	Program does not work. No results are produced. Unable to apply numerical methods learned to solve the problem.	8	
					TOTAL	15	

Note to students: Please print out and attach this appendix together with the submission of coursework